Verification Summary

How the volatility surface tool was checked

Kevin Williams | October 25, 2026

I checked the tool at four levels, each against an answer known in advance or computed by an unrelated program. First, the pricing formula passes the Black-Scholes equation and put-call parity checks in MATLAB's symbolic engine. Second, the solver turns known volatilities into prices and back again: 142 round trips came back with a worst error of 3×10^{-7}, and impossible prices are rejected instead of "solved". Third, the interpolated surface matches a surface with a known formula at points it was never given, and it leaves blank areas blank instead of filling in zeros. Finally, on a real AAPL chain (140 options, 8 expiries) MATLAB, the OCaml engine and an independent Python solver all agree to within 5×10^{-7}, which is OCaml's own six-decimal rounding.

See *Verification Report* and *Verification Process* in this folder for the full evidence and procedure.